

How to scan for vulnerabilities with GitHub Security Lab's open source AI-powered framework

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Content

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[Man Yue Mo](#) & [Peter Stöckli](#)

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For the last few months, we've been using the [GitHub Security Lab Taskflow Agent](#) along with a new set of auditing taskflows that specialize in finding web security vulnerabilities. They also turn out to be very successful at finding high-impact vulnerabilities in open source projects.

As security researchers, we're used to losing time on possible vulnerabilities that turn out to be unexploitable, but with these new taskflows, we can now spend more of our time on manually verifying the results and sending out reports. Furthermore, the severity of the vulnerabilities that we're reporting is uniformly high. Many of them are authorization bypasses or information disclosure vulnerabilities that allow one user to login as somebody else or to access the private data of another user.

Using these taskflows, we've reported more than 80 vulnerabilities so far. At the time of writing, approximately 20 of them have already been disclosed. And we're continually updating our [advisories page](#) when new vulnerabilities are disclosed. In this blog post, we'll show a few concrete examples of high-impact vulnerabilities that are found by these taskflows, like accessing personally identifiable information (PII) in shopping carts of ecommerce applications or signing in with *any* password into a chat application.

We'll also explain how the taskflows work, so you can learn how to write your own. The security community moves faster when it shares knowledge, which is why we've made the framework open source and easy to run on your own project. The more teams using and contributing to it, the faster we collectively eliminate vulnerabilities.

How to run the taskflows on your own project

Want to get started right away? The taskflows are open source and easy to run yourself! Please note: A [GitHub Copilot](#) license is required, and the prompts will use premium model requests. (Note that running the taskflows can result in many tool calls, which can easily consume a large amount of quota.)

- Go to the [seclab-taskflows](#) repository and start a codespace.
- Wait a few minutes for the codespace to initialize.
- In the terminal, run `./scripts/audit/run_audit.sh myorg/myrepo`

It might take an hour or two to finish on a medium-sized repository. When it finishes, it'll open an SQLite viewer with the results. Open the "audit_results" table and look for rows with a check-mark in the "has_vulnerability" column.

| **Tip:** Due to the non-deterministic nature of LLMs, it is worthwhile to perform multiple runs of these audit taskflows on the same codebase. In certain cases, a second run can lead to entirely different results. In addition to this, you might perform those two runs using different models (e.g., the first using GPT 5.2 and the second using Claude Opus 4.6). | |

The taskflows also work on private repos, but you'll need to modify the codespace configuration to do so because it won't allow access to your private repos by default.

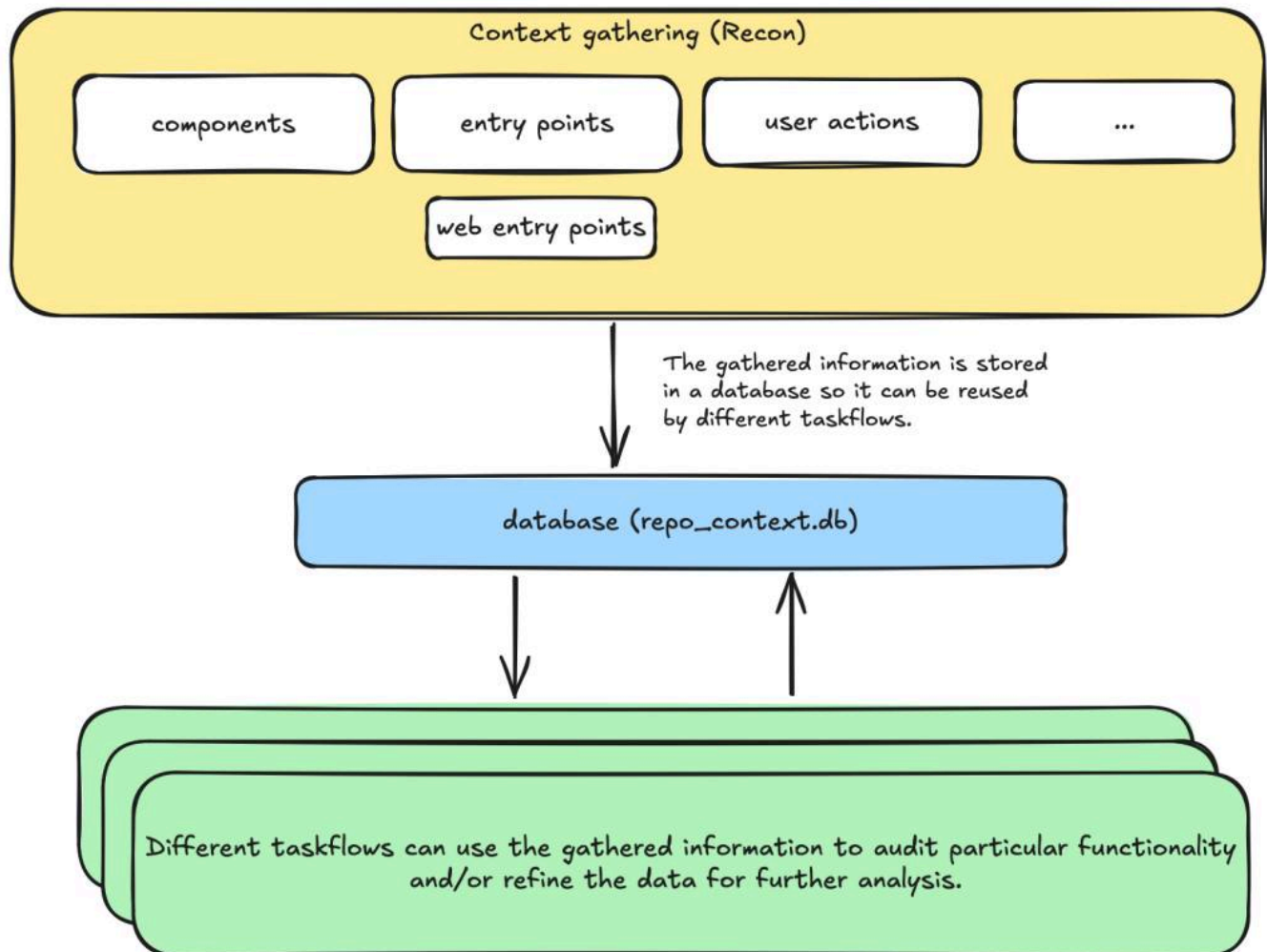
Introduction to taskflows

Taskflows are YAML files that describe a series of tasks that we want to do with an LLM. With them, we can write prompts to complete different tasks and have tasks that depend on each other. The [seclab-taskflow-agent](#) framework takes care of running the tasks sequentially and passing the results from one task to the next.

For example, when auditing a repository, we first divide the repository into different components according to their functionalities. Then, for each component, we may want to collect some information such as entry points where it takes untrusted input from, intended privilege, and purposes of the component, etc. These results are then stored in a database to provide the context for subsequent tasks.

Based on the context data, we can then create different auditing tasks. Currently, we have a task that suggests some generic issues for each component and another task that carefully audits each suggested issue. However, it's also possible to create other tasks, such as tasks with specific focus on a certain type of issue.

These become a list of tasks we specify in a taskflow file.



We use tasks instead of one big prompt because LLMs have limited context windows, and complex, multi-step tasks are often not completed properly. For example, some steps can be left out. Even though some LLMs have larger context windows, we find that taskflows are still useful in providing a way for us to control and debug the tasks, as well as for accomplishing bigger and more complex projects.

The seclab-taskflow-agent can also run the same task across many components asynchronously (like a for loop). During audits, we often reuse the same prompt and task for every component, varying only the details. The seclab-taskflow-agent lets us define templated prompts, iterate through components, and substitute component-specific details as it runs.

Taskflows for general security code audits

After using seclab-taskflow-agent to [triage CodeQL alerts](#), we decided we didn't want to restrict ourselves to specific types of vulnerabilities and started to explore using the

framework for more general security auditing. The main challenge in giving LLMs more freedom is the possibility of hallucinations and an increase in false positives. After all, the success with triaging CodeQL alerts was partly due to the fact that we gave the LLM a very strict and well-defined set of instructions and criteria, so the results could be verified at each stage to see if the instructions were followed.

So our goal here was to find a good way to allow the LLM the freedom to look for different types of vulnerabilities while keeping hallucinations under control.

We're going to show how we used agent taskflows to discover high-impact vulnerabilities with high true positive rate using just taskflow design and prompt engineering.

General taskflow design

To minimize hallucinations and false positives at the taskflow design level, our taskflow starts with a threat modelling stage, where a repository is divided into different components based on functionalities and various information, such as entry points, and the intended use of each component is collected. This information helps us to determine the security boundary of each component and how much exposure it has to untrusted input.

The information collected through the threat modelling stage is then used to determine the security boundary of each component and to decide what should be considered a security issue. For example, a command injection in a CLI tool with functionality designed to execute any user input script may be a bug but not a security vulnerability, as an attacker able to inject a command using the CLI tool can already execute any script.

At the level of prompts, the intended use and security boundary that is discovered is then used in the prompts to provide [strict guidelines](#) as to whether an issue found should be considered a vulnerability or not.

You need to take into account of the intention and threat model of the component in component notes to determine **if** an issue is a valid security issue or **if** it is an intended functionality. You can fetch entry points, web entry points and user actions to help you determine the intended usage of the component.

Asking an LLM something as vague as looking for any type of vulnerability anywhere in the code base would give poor results with many hallucinated issues. Ideally, we'd like to simulate the triage environment where we have some potential issues as the starting point of analysis and ask the LLM to apply rigorous criteria to determine whether the potential issue is valid or not.

To bootstrap this process, we break the auditing task into two steps.

- First, we ask the LLM to go through each component of the repository and suggest types of vulnerabilities that are more likely to appear in the component.
- These suggestions are then passed to another task, where they will be audited according to rigorous criteria.

In this setup, the suggestions from the first step act as some inaccurate vulnerability alerts flagged by an “external tool,” while the second step serves as a triage step. While this may look like a self-validating process—by breaking it down into two steps, each with a fresh context and different prompts—the second step is able to provide an accurate assessment of suggestions.

We’ll now go through these tasks in detail.

Threat modeling stage

When triaging alerts flagged by automatic code scanning tools, we found that a large proportion of false positives is the result of improper threat modeling. Most static analysis tools do not take into account the intended usage and security boundary of the source code and often give results that have no security implications. For example, in a reverse proxy application, many SSRF (server-side request forgery) vulnerabilities flagged by automated tools are likely to fall within the intended use of the application, while some web services used, for example, in continuous integration pipelines are designed to execute arbitrary code and scripts within a sandboxed environment. Remote code execution vulnerabilities in these applications without a sandboxed escape are generally not considered a security risk.

Given these caveats, it pays to first go through the source code to get an understanding of the functionalities and intended purpose of code. We divide this process into the following tasks:

- **Identify applications:** A GitHub repository is an imperfect boundary for auditing: It may be a single component within a larger system or contain multiple components, so it’s worth identifying and auditing each component separately to match distinct security boundaries and keep scope manageable. We do this with the `identify_applications` taskflow, which asks the LLM to inspect the repository’s source code and documentation and divide it into components by functionality.
- **Identify entry points:** We identify how each entry point is exposed to untrusted inputs to better gauge risk and anticipate likely vulnerabilities. Because “untrusted input” varies significantly between libraries and applications, we provide separate guidelines for each case.
- *Identify web entry points:* **This is an extra step** to gather further information about entry points in the application and append information that is specific to web application entry points such as noting the HTTP method and paths that are required to access a certain endpoint.
- **Identify user actions:** We have the LLM review the code and identify what functionality a user can access under normal operation. This clarifies the user’s baseline privileges, helps assess whether vulnerabilities could enable privilege gains, and informs the component’s security boundary and threat model, with separate instructions depending on whether the component is a library or an application.

At each of the above steps, information gathered about the repository is stored in a database. This includes components in the repository, their entry points, web entry points,

and intended usage. This information is then available for use in the next stage.

Issue suggestion stage

At this stage, we instruct the LLM to suggest some types of vulnerabilities, or a general area of high security risk for each component [based on the information](#) about the entry point and intended use of the component gathered from the previous step. In particular, we put emphasis on the intended usage of the component and its risk from untrusted input:

Base your decision on:

- Is [this](#) component likely to take untrusted user input? [For](#) example, remote web request or IPC, RPC calls?
- What is the intended purpose of [this](#) component and its functionality? Does it allow high privileged action?

Is it intended to provide such functionalities [for](#) all user? Or is there complex access control logic involved?

- The component itself may also have its own ``README.md`` (or a subdirectory of it may have a ``README.md``). Take a look at those files to help understand the functionality of the component.

We also explicitly instruct the LLM to not suggest issues that are of low severity or are generally considered non-security issues.

However, you should still take care not to include issues that are of low severity or requires unrealistic attack scenario such as misconfiguration or an already compromised system.

In general, we keep this stage relatively free of restrictions and allow the LLM freedom to explore and suggest different types of vulnerabilities and potential security issues. The idea is to have a reasonable set of focus areas and vulnerability types for the actual auditing task to use as a starting point.

One problem we ran into was that the LLM would sometimes start auditing the issues that it suggested, which would defeat the purpose of the brainstorming phase. To prevent this, we instructed the LLM to [not audit](#) the issues.

Issue audit stage

This is the final stage of the taskflows. Once we've gathered all the information we need about the repository and have suggested some types of vulnerabilities and security risks to focus on, the taskflow goes through each suggested issue and audits them by going through the source code. At this stage, the task starts with fresh context to scrutinize the issues suggested from the previous stage. The suggestions are considered to be [invalidated](#), and this taskflow is instructed to verify these issues:

The issues suggested have not been properly verified and are only suggested because they are common issues in these types of application. Your task is to audit the source code to check **if this** type of issues is present.

To avoid the LLM coming up with issues that are non-security related in the context of the component, we once again emphasize that intended usage must be taken into consideration.

You need to take into account of the intention and threat model of the component in component notes to determine **if** an issue is a valid security issue or **if** it is an intended functionality.

To avoid the LLM hallucinating issues that are unrealistic, we also instruct it to provide a concrete and realistic attack scenario and to only consider issues that stem from errors in the source code:

Do not consider scenarios where authentication is bypassed via stolen credential etc. We only consider situations that are achievable **from** within the source code itself.

...

If you believe there is a vulnerability, then you must include a realistic attack scenario, with details of all the file and line included, and also what an attacker can gain by exploiting the vulnerability. Only consider the issue a vulnerability **if** an attacker can gain privilege by performing an action that is not intended by the component.

To further reduce hallucinations, we also instruct the LLM to provide concrete evidence from the source code, with file path and line information:

Keep a record of the audit notes, be sure to include all relevant file path and line number. Just stating an end point, e.g. ``IDOR in user update/delete endpoints (PUT /user/:id)`` is not sufficient. I need to have the file and line number.

Finally, we also instruct the LLM that it is possible that there is no vulnerability in the component and that it should not make things up:

Remember, the issues suggested are only speculation and there may not be a vulnerability at all and it is ok to conclude that there is no security issue.

The emphasis of this stage is to provide accurate results while following strict guidelines—and to provide concrete evidence of the findings. With all these strict instructions in place, the LLM indeed rejects many unrealistic and unexploitable suggestions with very few hallucinations.

The first prototype was designed with hallucination prevention as a priority, which raised a question: Would it become too conservative, rejecting most vulnerability candidates and failing to surface real issues?

The answer is clear after we ran the taskflow on a few repositories.

Three examples of vulnerabilities found by the taskflows

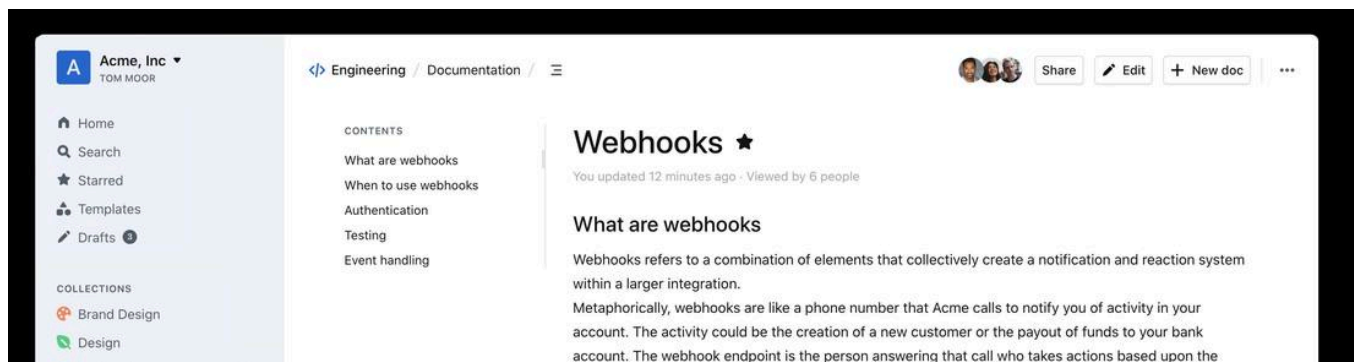
In this section, we'll show three examples of vulnerabilities that were found by the taskflows and that have already been disclosed. In total, we have found and reported over 80 vulnerabilities so far. We publish all disclosed vulnerabilities on our [advisories page](#).

Privilege escalation in Outline (CVE-2025-64487)

Our information-gathering taskflows are optimized toward web applications, which is why we first pointed our audit taskflows to a collaborative web application called [Outline](#).

Outline is a multi-user collaboration suite with properties we were especially interested in:

- Documents have owners and different visibility, with permissions per users and teams.
- Access rules like that are hard to analyze with a Static Application Security Testing (SAST) tool, since they use custom access mechanisms and existing SAST tools typically don't know what actions a normal "user" should be able to perform.
- Such permission schemes are often also hard to analyze for humans by only reading the source code (if you didn't create the scheme yourself, that is).



And success: Our taskflows found a bug in the authorization logic on the very first run!

The notes in the audit results read like this:

Audit target: Improper membership management authorization in component server (backend API) of outline/outline (component id 2).

Summary conclusion: A real privilege escalation vulnerability exists. The document group membership modification endpoints (documents.add_group, documents.remove_group) authorize with the weaker \"update\" permission instead of the stronger \"manageUsers\" permission that is required **for** user membership changes. Because \"update\" can be satisfied by having only a ReadWrite membership on the document, a non-admin document collaborator can grant (or revoke) group memberships – including granting Admin permission – thereby escalating their own privileges (**if** they are in the added group) and those of other group members. **This** allows actions (manageUsers, archive, **delete**, etc.) that were not intended **for** a mere ReadWrite collaborator.

Reading the TypeScript-based source code and verifying this finding on a test instance revealed that it was exploitable exactly as described. In addition, the described steps to exploit this vulnerability were on point:

Prerequisites:

- Attacker is a normal team member (not admin), not a guest, with direct ReadWrite membership on Document D (or via a group that grants ReadWrite) but NOT Admin.
- Attacker is a member of an existing group G in the same team (they **do** not need to be an admin of G; group read access is sufficient per group policy).

Steps:

1. Attacker calls POST documents.add_group (server/routes/api/documents/documents.ts lines 1875-1926) with body:

```
{  
  "id": "<document-D-id>",  
  "groupId": "<group-G-id>",  
  "permission": "admin"  
}
```

2. Authorization path:

- Line 1896: authorize(user, "update", document) succeeds because attacker has ReadWrite membership (document.ts lines 96-99 allow update).
- Line 1897: authorize(user, "read", group) succeeds **for** any non-guest same-team user (group.ts lines 27-33).

No \"manageUsers\" check occurs.

3. Code creates or updates GroupMembership with permission Admin (lines 1899-1919).

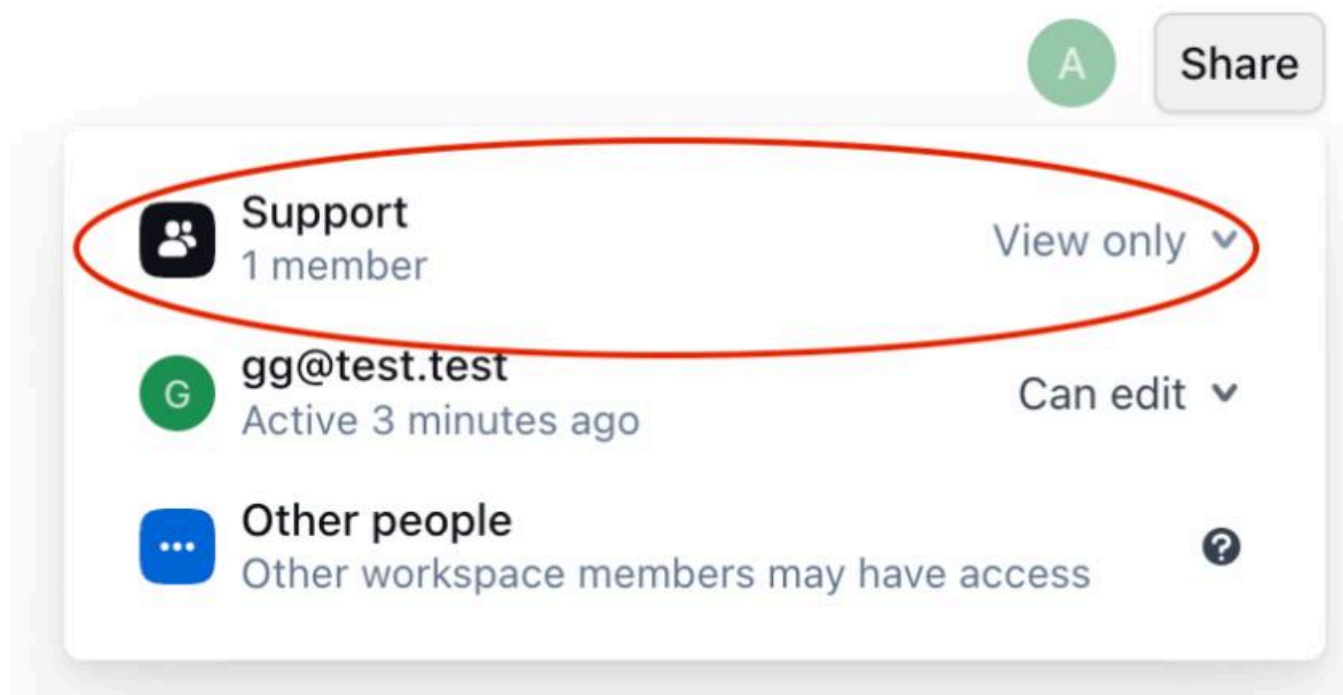
4. Because attacker is a member of group G, their effective document permission (via groupMembership) now includes DocumentPermission.Admin.

5. With Admin membership, attacker now satisfies includesMembership(Admin) used in:

- manageUsers (document.ts lines 123-134) enabling adding/removing arbitrary users via documents.add_user / documents.remove_user (lines 1747-1827, 1830-1872).
- archive/unarchive/**delete** (document.ts archive policy lines 241-252; **delete** lines 198-208) enabling content integrity impact.
- duplicate, move, other admin-like abilities (e.g., duplicate policy lines 136-153; move lines 155-170) beyond original ReadWrite scope.

Using these instructions, a low-privileged user could add arbitrary groups to a document that the user was only allowed to update (the user not being in the possession of the “manageUsers” permission that was typically required for such changes).

In this sample, the group “Support” was added to the document by the low-privileged user named “gg.”



The Outline project fixed this and another issue we reported within three days! ([Repo advisory](#))

The shopping cartocalypse (CVE-2025-15033, CVE-2026-25758)

We didn't realize what systematic issues we'd uncover in the cart logic of ecommerce applications until we pointed our taskflows at the first online shop in our list. In the PHP-based WooCommerce project, the taskflows promptly found a way for normally signed-in shop users to view all guest orders—including personally identifiable information (including names, addresses, and phone numbers). After we reported this, Automattic (the company behind WooCommerce) quickly released an update ([CVE-2025-15033](#)) and accompanying [blog post](#).

Intrigued by that vulnerability, we've added additional ecommerce applications to our list of applications to be audited by our agent. And sure enough, we found more vulnerabilities. The popular Ruby-based [Spree commerce](#) application contained two similar vulnerabilities ([CVE-2026-25758](#) and [CVE-2026-25757](#)). The more critical one allowed unauthenticated users to simply enumerate the addresses (and phone numbers) of all guest orders by more or less incrementing a sequential number.

In this screenshot, the attacker "test66" linked their session to an existing address of a guest user, thus being able to view the full address and phone number.

Contact information

Already have an account? [Login](#)

Email
test66@test.test

- ☐ Email me about new products, sales, and more. You can unsubscribe at any time.
- ☐ Create an account for faster future purchases, order tracking, and more.

Shipping Address

Country

France

First Name

Richard

Last Name

Address

Address (contd.)

Address (contd.)

City

Zip Code

Phone

1

Long Sleeve T Shirt

Color: Grey, Size: M

\$81.99

Add promo code

Apply

Subtotal:

\$81.99

Shipping:

Calculated at next step

Total

USD **\$86.99**

Our bug-hunting spree didn't stop with Spree. Our taskflows uncovered similar issues in two additional ecommerce applications.

These authorization logic bugs had been undiscovered for years.

Signing in to Rocket.Chat using *any* password (CVE-2026-28514)

(This is not what passwordless authentication should look like!)

Every so often you can't believe your eyes. This finding reported by our taskflows in Rocket.Chat was one of those moments.

When your agent comes back with a note like this:

VULNERABILITY: password authentication bypass in account-service allows logging in as any user with a password set.

You might find it hard to believe at first.

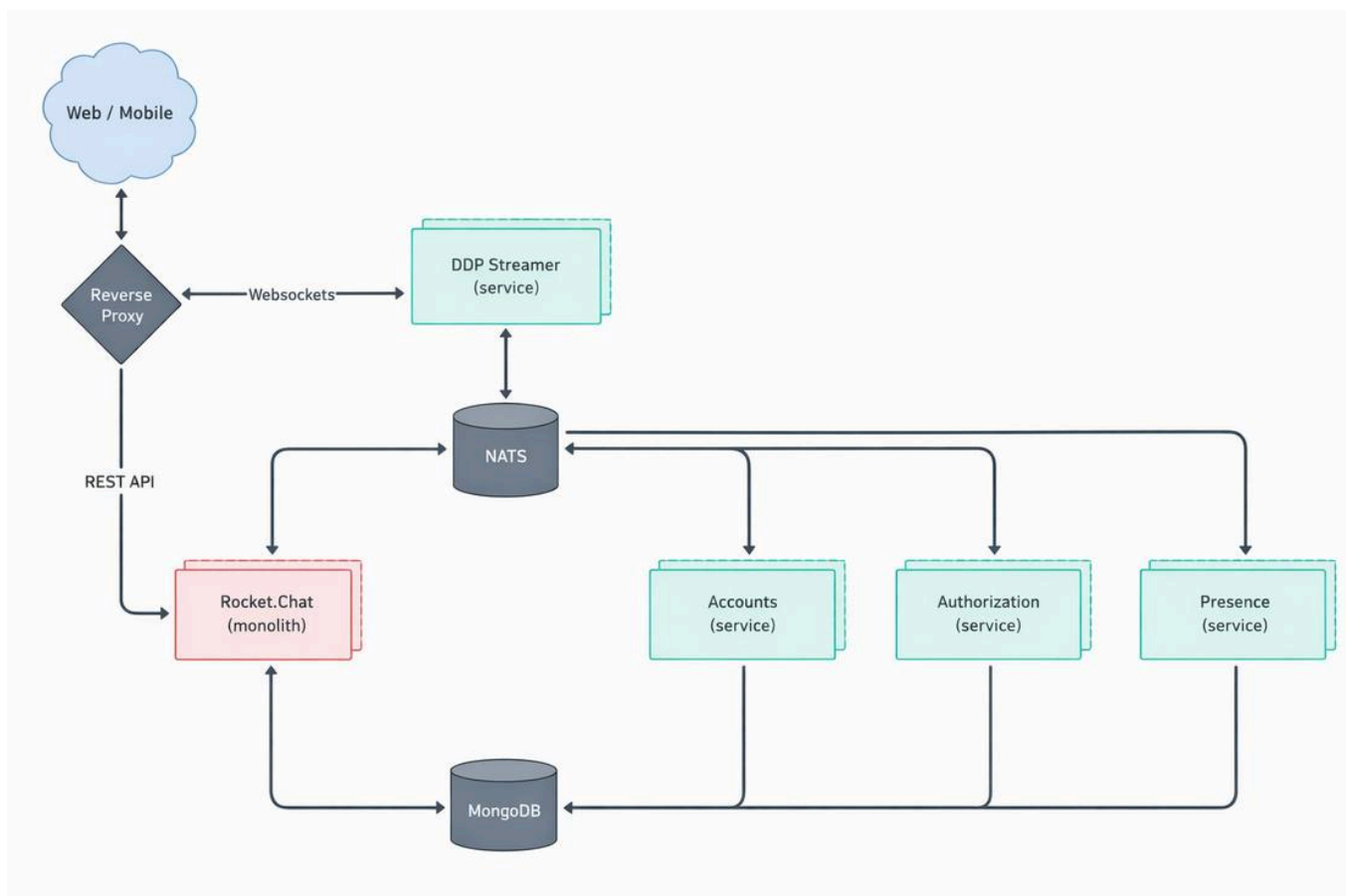
When you then continue reading the output:

Root cause:

- ee/apps/account-service/src/lib/utils.ts:60-61: `validatePassword` returns `Promise<boolean>` (`bcrypt.compare(...)`).
- ee/apps/account-service/src/lib/loginViaUsername.ts:18-21: `const valid = user.services?.password?.bcrypt && validatePassword(password, user.services.password.bcrypt);` but does NOT `await` the Promise; since a Promise is truthy, `if (!valid) return false;` is never triggered when bcrypt hash exists.
- ee/apps/account-service/src/lib/loginViaUsername.ts:23-35: proceeds to mint a **new** login token and saves it, returning `{ uid, token, hashedToken, tokenExpires }`.

It might make more sense, but you're still not convinced.

It turns out the suspected finding is in the micro-services based setup of Rocket.Chat. In that particular setup, Rocket.Chat exposes its user account service via its DDP Streamer service.



*Rocket.Chat's microservices deployment Copyright Rocket.Chat. (This architecture diagram is from** [Rocket.Chat's documentation](#)**.)**

Once our Rocket.Chat test setup was working properly, we had to write proof of concept code to exploit this potential vulnerability. The notes of the agent already contained the JSON construct that we could use to connect to the endpoint using Meteor's DDP protocol. We connected to the WebSocket endpoint for the DDP streamer service, and yes: It was truly possible to login into the exposed Rocket.Chat DDP service using any password. Once

signed in, it was also possible to perform other operations such as connecting to arbitrary chat channels and listening on them for messages sent to those channels.

Here we received the message “HELLO WORLD!!!” while listening on the “General” channel.

```
Message: {"msg":"connected","session":"f45b34c0-ed2b-11f0-bd8a-37b23a4efb49"}
Message: {"msg":"result","id":"newid2","result":{"id":"WfKb7F2xzFnM3pnpe","token":"50e62751-ba18-4f62-9592-5c38bd4b46c6","tokenExpires":{"$date":1775719348898},"type":"password"}}}
Message: {"msg":"updated","methods":["newid2"]}
Message: {"msg":"ready","subs":["newid3"]}
Message: {"msg":"added","collection":"users","id":"WfKb7F2xzFnM3pnpe","fields":{"_id":"WfKb7F2xzFnM3pnpe","services":{"email2fa":{"enabled":true}},"emails":{"address":"user1@test.test","verified":false},"roles":["user"],"status":"online","active":true,"name":"user1","_updatedAt":{"$date":1767943348898},"username":"user1","statusConnection":"online","utcOffset":1,"statusDefault":"online"}}}
Message: {"msg":"changed","collection":"stream-room-messages","id":"id","fields":{"eventName":"GENERAL","args":{"_id":"9f5krYfLQrgDdPscQ","rid":"GENERAL","msg":"cf","ts":{"$date":1767943365664},"u":{"_id":"WfKb7F2xzFnM3pnpe","username":"user1","name":"user1","_updatedAt":{"$date":1767943365678},"urls":[],"mentions":[],"channels":[],"md":{"type":"PARAGRAPH","value":{"type":"PLAIN_TEXT","value":"cf"}}}}}}}
Message: {"msg":"ping"}
Message: {"msg":"changed","collection":"stream-room-messages","id":"id","fields":{"eventName":"GENERAL","args":{"_id":"7gibjcMjPgiwQysD","rid":"GENERAL","msg":"HELLO WORLD!!!","ts":{"$date":1767943387306},"u":{"_id":"WfKb7F2xzFnM3pnpe","username":"user1","name":"user1","_updatedAt":{"$date":1767943387319},"urls":[],"mentions":[],"channels":[],"md":{"type":"PARAGRAPH","value":{"type":"PLAIN_TEXT","value":"HELLO WORLD!!!"}}}}}}}
```

The technical details of this issue are interesting (and scary as well). Rocket.Chat, primarily a TypeScript-based web application, uses bcrypt to store local user passwords. The `bcrypt.compare` function (used to compare a password against its stored hash) returns a [Promise](#)—a fact that is reflected in Rocket.Chat’s own `validatePassword` function, which returns `Promise<boolean>`:

```
export const validatePassword = (password: string, bcryptPassword: string):
Promise<boolean> =>
    bcrypt.compare(getPassword(password), bcryptPassword);
```

However, when that function was used, the value of the Promise was not settled (e.g. by adding an `await` keyword in front of `validatePassword`):

```
const valid = user.services?.password?.bcrypt && validatePassword(password,
user.services.password.bcrypt);

if (!valid) {
    return false;
}
```

This led to the result of `validatePassword` being ANDed with `true`. Since a returned Promise is always “truthy” speaking in JavaScript terms, the boolean `valid` subsequently was always true when a user had a bcrypt password set.

Severity aside, it’s fascinating that the LLM was able to pick up this rather subtle bug, follow it through multiple files, and arrive at the correct conclusion.

What we learned

After running the taskflows over 40 repositories—mostly multi-user web applications—the LLM suggested 1,003 issues (potential vulnerabilities).

After the audit stage, 139 were marked as having vulnerabilities, meaning that the LLM decided they were exploitable. After deduplicating the issues—duplicates happen because each repository is run a couple of times on average and the results are aggregated—we end up with 91 vulnerabilities, which we decided to manually inspect before reporting.

- We rejected 20 (22%) results as FP: False Positives that we couldn't reproduce manually.
- We rejected 52 (57%) results as low severity: Issues that have very limited potential impact (e.g., blind SSRF with only a HTTP status code returned, issues that require malicious admin during installation stage, etc.).
- We kept only 19 (21%) results that we considered vulnerabilities impactful enough to report, all serious vulnerabilities with the majority having a high or critical severity (e.g., vulnerabilities that can be triggered without specific requirements with impact to confidentiality or integrity, such as disclosure of personal data, overwriting of system settings, account takeover, etc.).

This data was collected using `gpt-5.x` as the model for code analysis and audit tasks.

Note that we have run the taskflows on more repositories since this data was collected, so this table does not represent all the data we've collected and all vulnerabilities we've reported.

Issue category All Has vulnerability Vulnerability rate IDOR/Access control issue 241 38 15.8% XSS 131 17 13.0% CSRF 110 17 15.5% Authentication issue 91 15 16.5% Security misconfiguration 75 13 17.3% Path traversal 61 10 16.4% SSRF 45 7 15.6% Command injection 39 5 12.8% Remote code execution 24 1 4.2% Business logic issue 24 6 25.0% Template injection 24 1 4.2% File upload handling issues (excludes path traversal) 18 2 11.1% Insecure deserialization 17 0 0.0% Open redirect 16 0 0.0% SQL injection 9 0 0.0% Sensitive data exposure 8 0 0.0% XXE 4 0 0.0% Memory safety 3 0 0.0% Others 66 7 10.6%
--

If we divide the findings into two rough categories—logical issues (IDOR, authentication, security misconfiguration, business logic issues, sensitive data exposure) and technical issues (XSS, CSRF, path traversal, SSRF, command injection, remote code execution, template injection, file upload issues, insecure deserialization, open redirect, SQL injection, XXE, memory safety)—we get 439 logical issues and 501 technical issues. Although more technical issues were suggested, the difference isn't significant because some broad categories (such as remote code execution and file upload issues) can also involve logical issues depending on the attacker scenario.

There are only three suggested issues that concern memory safety. This isn't too surprising, given the majority of the repositories tested are written in memory-safe languages. But we also suspect that the current taskflows may not be very efficient in finding memory-safety issues, especially when comparing to other automated tools such as fuzzers. This is an interesting area that can be improved by creating more specific taskflows and making more tools, like fuzzers, available to the LLM.

This data led us to the following observations.

LLMs are particularly good at finding logic bugs

What stands out from the data is the 25% rate of “Business logic issue” and the large amount of IDOR issues. In fact, the total number of IDOR issues flagged as vulnerable is more than the next two categories combined (XSS and CSRF). Overall, we get the impression that the LLM does an excellent job of understanding the code space and following the control flow, while taking into account the access control model and intended usage of the application, which is more or less what we’d expect from LLMs that excel in tasks like code reviews. This also makes it great for finding logic bugs that are difficult to find with traditional tools.

LLMs are good at rejecting low-severity issues and false positives

Curiously, none of the false positives are what we’d consider to be hallucinations. All the reports, including the false positives, have sound evidence backing them up, and we were able to follow through the report to locate the endpoints and apply the suggested payload. Many of the false positives are due to more complex circumstances beyond what is available in the code, such as browser mitigations for XSS issues mentioned above or what we would consider as genuine mistakes that a human auditor is also likely to make. For example, when multiple layers of authentications are in place, the LLM could sometimes miss out some of the checks, resulting in false positives.

We have since tested more repositories with more vulnerabilities reported, but the ratio between vulnerabilities and repositories remains roughly the same.

To demonstrate the extensibility of taskflows and how extra information can be incorporated into the taskflows, we created a new [taskflow](#) to run after the audit stage, which incorporates our new-found knowledge to filter out low-severity vulnerabilities. We found that the taskflow can filter out roughly 50% of the low-severity vulnerabilities with a couple of borderline vulnerabilities that we reported also getting marked as low severity. The taskflow and the prompt can be adjusted to fit the user’s own preference, but for us, we’re happy to make it more inclusive so we don’t miss out on anything impactful.

LLMs are good at threat modeling

The LLM performs well in threat modeling in general. During the experiment, we tested it on a number of applications with different threat models, such as desktop applications, multi-tenant web applications, applications that are designed to run code in sandbox environments (code injection by design), and reverse proxy applications (applications where SSRF-like behavior is intended). The taskflow is able to take into account the intended usage of these applications and make sound decisions. The taskflow struggles most with threat modelling of desktop applications, as it is often unclear whether other processes running on the user’s desktop should be considered trusted or not.

We've also observed some remarkable reasoning by the LLM that excludes issues with no privilege gains. For example, in one case, the LLM noticed that while there are inconsistencies in access control, the issue does not give the attacker any advantages over a manual copy and paste action:

Security impact assessment:

A user possessing only read access to a document (no update rights) can duplicate it provided they also have updateDocument rights on the destination collection. **This** allows creation of a **new** editable copy of content they could already read. **This** does NOT grant additional access to other documents nor bypass protections on the original; any user with read access could manually copy-paste the content into a **new** document they are permitted to create (creation generally allowed **for** non-guest, non-viewer members in ReadWrite collections per createDocument collection policy)

We've also seen some more sophisticated techniques that were used in the reasoning. For example, in one application that is running scripts in a sandboxed nodejs environment, the LLM suggested the following technique to escape the sandbox:

In Node's vm, passing any outer-realm **function** into a contextified sandbox leaks that **function's** outer-realm **Function** constructor through the ``constructor`` property. **From** inside the sandbox:

```
const F = console.log.constructor; // outer-realm Function
const hostProcess = F('return process')(); // host process object
// Bypass module allowlist via host dynamic import
const cp = await F('return import("node:child_process")')();
const out = cp.execSync('id').toString();
return [{ json: { out } }];
```

The presence of host functions (console.log, timers, require, RPC methods) is sufficient to obtain the host **Function** constructor and escape the sandbox. The allowlist in require-resolver is bypassed by constructing host-realm functions and **using** dynamic **import** of built-in modules (e.g., node:child_process), which does not **go** through the sandbox's custom require.

While the result turns out to be a false positive due to other mitigating factors, it demonstrates the LLM's technical knowledge.

Get involved!

The **taskflows** we used to find these vulnerabilities are open source and easy to run on your own project, so we hope you'll give them a try! We also want to encourage you to **write your own taskflows**. The results showcased in this blog post are just small examples of what's possible. There are other types of vulnerabilities to find, and there are other

security-related problems, like [triaging SAST results](#) or building development setups, which we think taskflows can help with. Let us know what you're building by [starting a discussion on our repo](#)!

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- [agentic AI](#)
- [CVE](#)
- [GitHub Security Lab](#)
- [open source](#)

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